Name of Student: Minisha Singh	**PRN: 25070122292**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 30	
TITLE: Write a JSP application to demonstrate CRUD operations.	

Problem Statement

Write a JSP application to demonstrate CRUD operations.

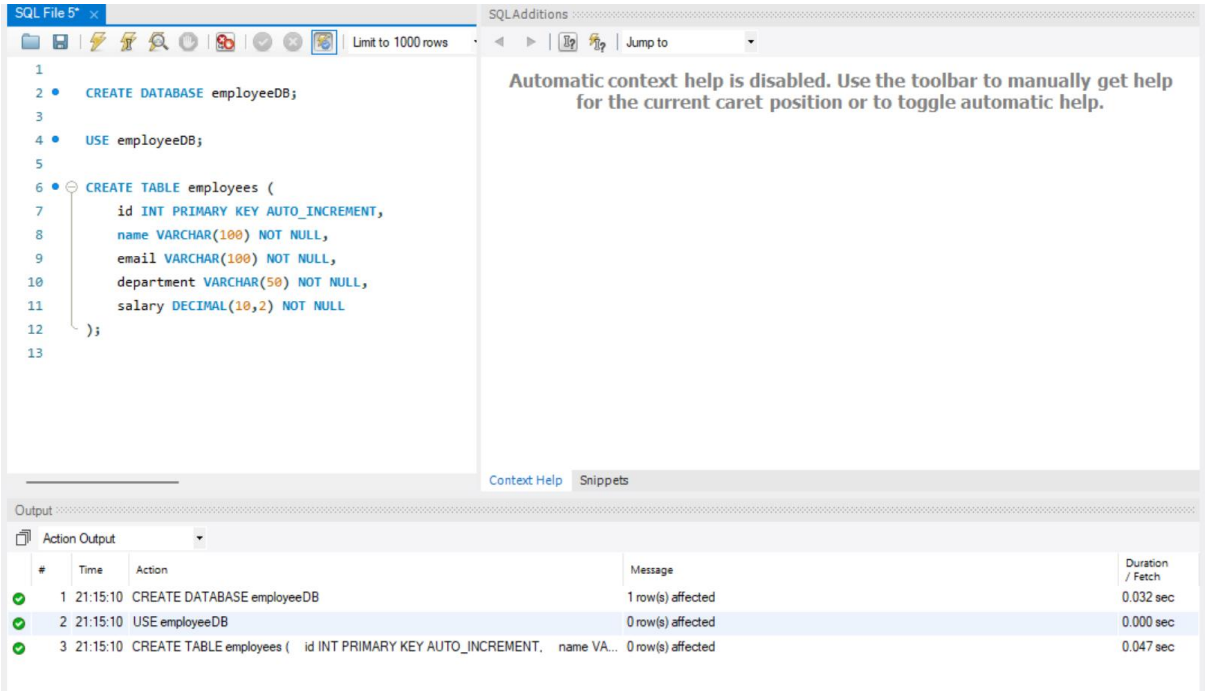
Learning Objectives

To implement Create, Read, Update, and Delete operations using JSP and a database.

Programme

1. Create a JSP-based application to add, update, delete, and view employee details

SQL Query:




```
CREATE DATABASE employeeDB;

USE employeeDB;

CREATE TABLE employees (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL,
    department VARCHAR(50) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);
```

Index.jsp Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Employee Management System</title>
</head>
<body>
    <h1>Employee Management System</h1>

    <h2>Add Employee</h2>
    <form action="addEmployee.jsp" method="post">
        Name: <input type="text" name="name" required><br><br>
        Email: <input type="email" name="email" required><br><br>
        Department: <input type="text" name="department" required><br><br>
        Salary: <input type="number" name="salary" step="0.01" min="0" required><br><br>
        <input type="submit" value="Add Employee">
    </form>

    <hr>

    <h2>Manage Employees</h2>
    <a href="viewEmployees.jsp">View / Update / Delete Employees</a>
</body>
</html>
```

addEmployee.jsp Code:

```
<%@ page import="java.sql.*" %>
<%
   String name = request.getParameter("name");
   String email = request.getParameter("email");
   String department = request.getParameter("department");
   String salary = request.getParameter("salary");

   Connection con = null;
   PreparedStatement ps = null;

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/employeeDB",
         "root",
         "PASSWORD"
      );

      String sql = "INSERT INTO employees (name, email, department,
salary) VALUES (?, ?, ?, ?)";
      ps = con.prepareStatement(sql);

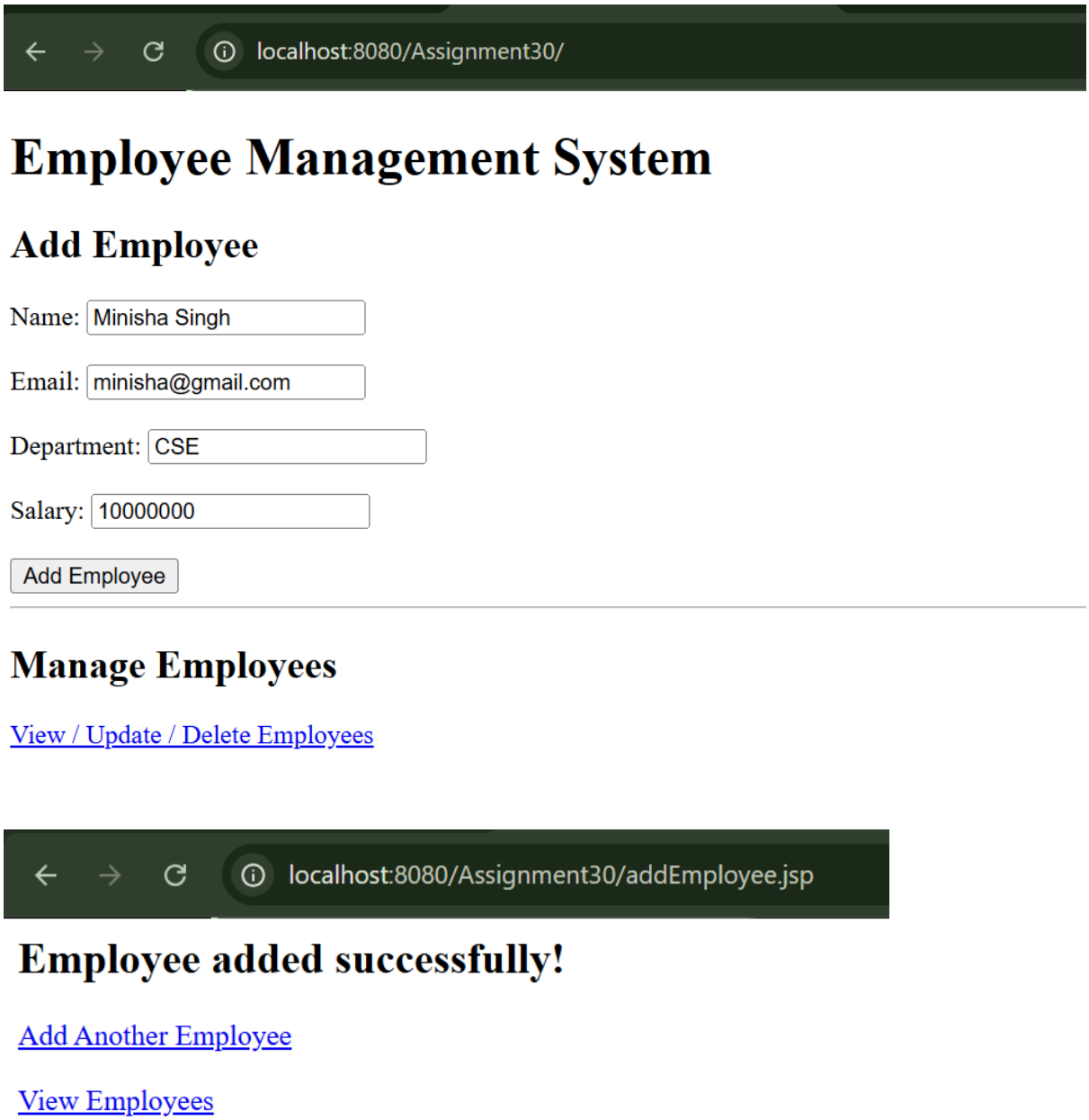
      ps.setString(1, name);
      ps.setString(2, email);
      ps.setString(3, department);
      ps.setDouble(4, Double.parseDouble(salary));

      int result = ps.executeUpdate();

      if (result > 0) {
%>
         <h2>Employee added successfully!</h2>
         <a href="index.jsp">Add Another Employee</a><br><br>
         <a href="viewEmployees.jsp">View Employees</a>
<%
      }
   } catch (Exception e) {
%>
      <h2>Error adding employee</h2>
      <p><%= e.getMessage() %></p>
      <a href="index.jsp">Go Back</a>
```

```
<%
   } finally {
      try { if (ps != null) ps.close(); } catch (Exception e) {}
      try { if (con != null) con.close(); } catch (Exception e) {}
   }
%>
</body>
</html>
```

Output:



viewEmployees.jsp Code:

```
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
   <title>View Employees</title>
</head>
<body>
   <h1>Employee Records</h1>

   <table border="1" cellpadding="10">
      <tr>
         <th>ID</th>
         <th>Name</th>
         <th>Email</th>
         <th>Department</th>
         <th>Salary</th>
         <th>Action</th>
      </tr>

      <%
         try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/employeeDB",
               "root",
               " PASSWORD"
            );

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM employees");

            while (rs.next()) {
      %>
      <tr>
         <td><%= rs.getInt("id") %></td>
         <td><%= rs.getString("name") %></td>
         <td><%= rs.getString("email") %></td>
         <td><%= rs.getString("department") %></td>
         <td><%= rs.getDouble("salary") %></td>
         <td>
```

```
            <a href="deleteEmployee.jsp?id=<%= rs.getInt("id") %>"
               onclick="return confirm('Delete this employee?')">
               Delete
            </a>
          </td>
        </tr>
        <%
            }
            rs.close();
            st.close();
            con.close();

        } catch (Exception e) {
        %>
          <tr>
            <td colspan="6">Error: <%= e.getMessage() %></td>
          </tr>
        <%
          }
        %>
    </table>

    <br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

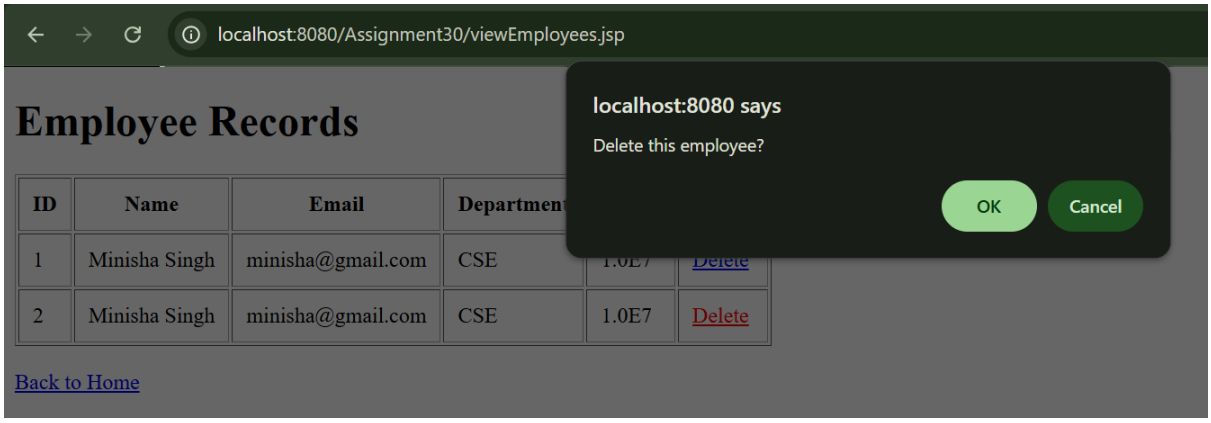
Output:



Employee Records

ID	Name	Email	Department	Salary	Action
1	Minisha Singh	minisha@gmail.com	CSE	1.0E7	Delete
2	Minisha Singh	minisha@gmail.com	CSE	1.0E7	Delete

Back to Home



deleteEmployee.jsp Code:

```
<%@ page import="java.sql.*" %>
<%
   String id = request.getParameter("id");

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/employeeDB",
         "root",
         " PASSWORD"
      );

      PreparedStatement ps = con.prepareStatement(
         "DELETE FROM employees WHERE id = ?"
      );

      ps.setInt(1, Integer.parseInt(id));

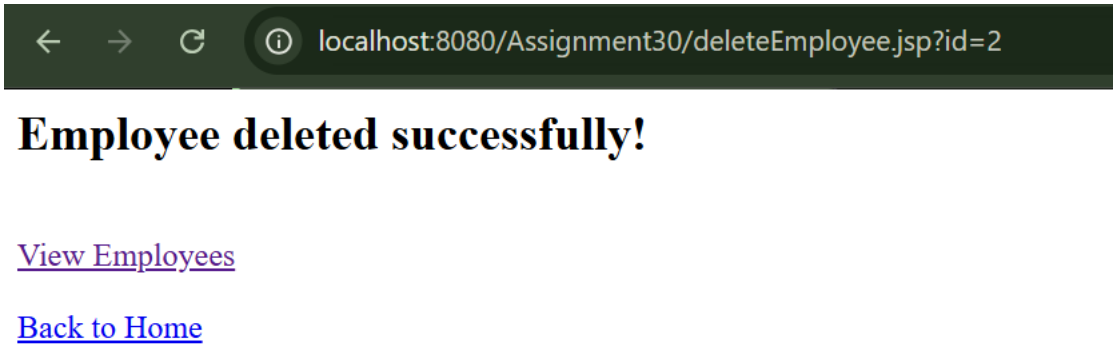
      int result = ps.executeUpdate();

      ps.close();
      con.close();

      if (result > 0) {
%>
         <h2>Employee deleted successfully!</h2>
<%
      } else {
%>
```

```
            <h2>Employee not found!</h2>
    <%
        }
    %>
        <br>
        <a href="viewEmployees.jsp">View Employees</a><br><br>
        <a href="index.jsp">Back to Home</a>
    <%
    } catch (Exception e) {
    %>
        <h2>Error deleting employee</h2>
        <p><%= e.getMessage() %></p>
        <a href="viewEmployees.jsp">Go Back</a>
    <%
    }
    %>
    </body>
    </html>
```

Output:



editEmployee.jsp Code:

```
<%@ page import="java.sql.*" %>
<html>
<body>
<h2>Update Employee Details</h2>

<%
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/employeeDB",
    "root",
    " PASSWORD"
```

```
);

int id = Integer.parseInt(request.getParameter("id"));

PreparedStatement ps = con.prepareStatement(
   "SELECT * FROM employees WHERE id=?"
);
ps.setInt(1, id);
ResultSet rs = ps.executeQuery();

if(rs.next()) {
%>

<form action="updateEmployee.jsp" method="post">
   <input type="hidden" name="id" value="<%= rs.getInt("id") %>">

   Name: <input type="text" name="name" value="<%= rs.getString("name")
%>" required><br><br>

   Email: <input type="email" name="email" value="<%= rs.getString("email")
%>" required><br><br>

   Department: <input type="text" name="department" value="<%=
rs.getString("department") %>" required><br><br>

   Salary: <input type="number" name="salary" step="0.01" value="<%=
rs.getDouble("salary") %>" required><br><br>

   <input type="submit" value="Update Employee">
</form>

<%
}
rs.close();
ps.close();
con.close();
%>

</body>
</html>
```

updateEmployee.jsp Code:

```
<%@ page import="java.sql.*" %>
<html>
<body>

<%
try {
   Class.forName("com.mysql.cj.jdbc.Driver");

   Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/employeeDB",
      "root",
      " PASSWORD"
   );

   int id = Integer.parseInt(request.getParameter("id"));
   String name = request.getParameter("name");
   String email = request.getParameter("email");
   String department = request.getParameter("department");
   double salary = Double.parseDouble(request.getParameter("salary"));

   PreparedStatement ps = con.prepareStatement(
      "UPDATE employees SET name=?, email=?, department=?, salary=?
WHERE id=?"
   );

   ps.setString(1, name);
   ps.setString(2, email);
   ps.setString(3, department);
   ps.setDouble(4, salary);
   ps.setInt(5, id);

   int result = ps.executeUpdate();

   if (result > 0) {
      out.println("<h2>Employee updated successfully!</h2>");
   } else {
      out.println("<h2>Employee not found!</h2>");
   }

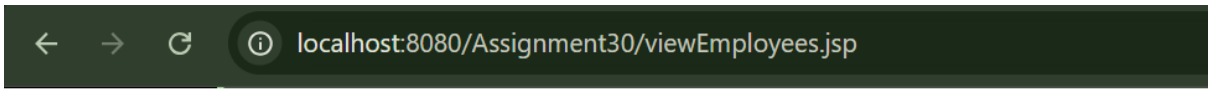
   ps.close();
   con.close();
```

```
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>

<br><br>
<a href="viewEmployees.jsp">View Employees</a><br><br>
<a href="index.jsp">Back to Home</a>

</body>
</html>
```

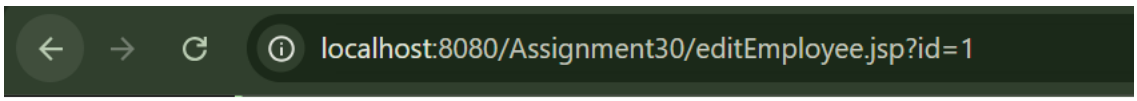
Output:



Employee Records

ID	Name	Email	Department	Salary	Action
1	Minisha Singh	minisha@gmail.com	CSE	1.0E7	Update | Delete

Back to Home



Update Employee Details

Name: Riya Sharma

Email: riya@gmail.com

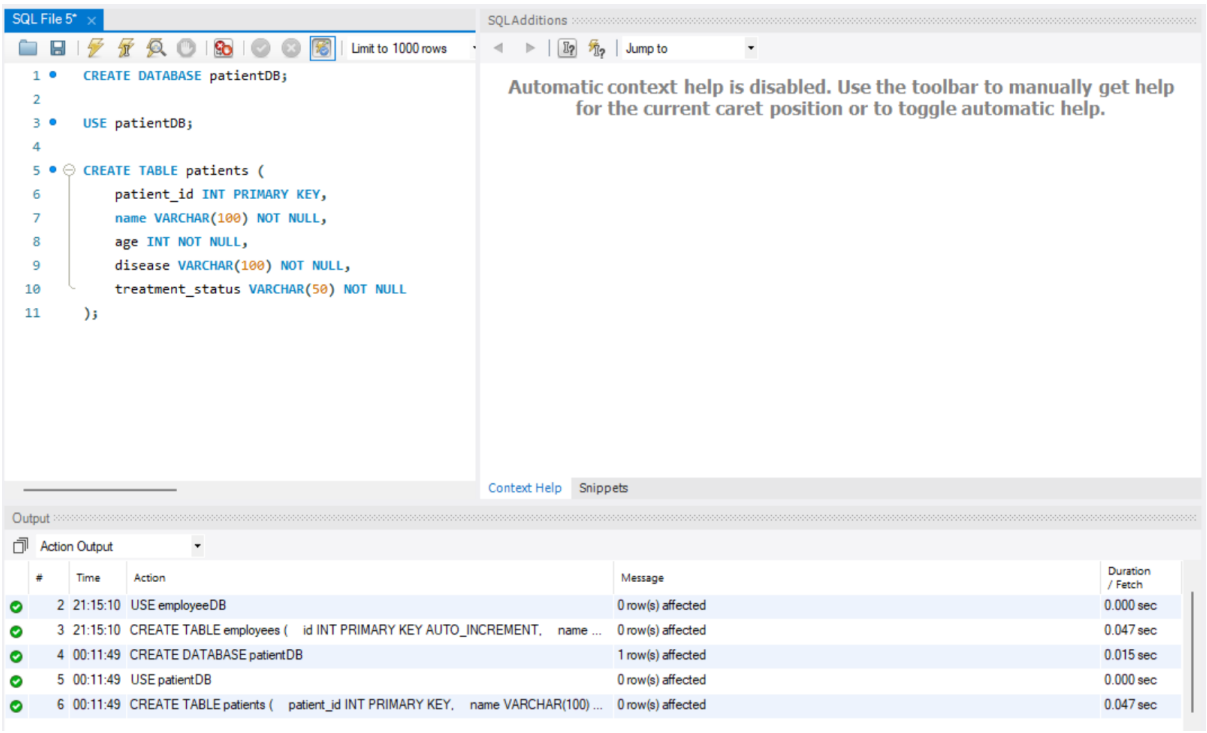
Department: CSE

Salary: 1.0E7

Update Employee

2. Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status.

SQL Query:




```
CREATE DATABASE patientDB;

USE patientDB;

CREATE TABLE patients (
    patient_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    age INT NOT NULL,
    disease VARCHAR(100) NOT NULL,
    treatment_status VARCHAR(50) NOT NULL
);
```

index.jsp Code:

```
<html>
<head>
    <title>Patient Management System</title>
</head>
<body>
    <h1>Patient Management System</h1>

    <h2>Add Patient</h2>

    <form action="addPatient.jsp" method="post">
        Patient ID: <input type="number" name="patient_id" required><br><br>
        Name: <input type="text" name="name" required><br><br>
        Age: <input type="number" name="age" min="0" required><br><br>
        Disease: <input type="text" name="disease" required><br><br>

        Treatment Status:
        <select name="treatment_status">
            <option value="Under Treatment">Under Treatment</option>
```

```
            <option value="Recovered">Recovered</option>
            <option value="Discharged">Discharged</option>
        </select><br><br>

        <input type="submit" value="Add Patient">
    </form>

    <hr>
    <a href="viewPatients.jsp">View / Update / Delete Patients</a>
</body>
</html>
```

addPatient.jsp Code:

```
<%@ page import="java.sql.*" %>
<html><body>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/patientDB",
        "root",
        " PASSWORD"
    );

    PreparedStatement ps = con.prepareStatement(
        "INSERT INTO patients VALUES (?, ?, ?, ?, ?)"
    );

    ps.setInt(1, Integer.parseInt(request.getParameter("patient_id")));
    ps.setString(2, request.getParameter("name"));
    ps.setInt(3, Integer.parseInt(request.getParameter("age")));
    ps.setString(4, request.getParameter("disease"));
    ps.setString(5, request.getParameter("treatment_status"));

    ps.executeUpdate();
    out.println("<h2>Patient added successfully!</h2>");

    ps.close();
    con.close();
} catch(Exception e) {
    out.println("Error: " + e.getMessage());
```

```
}
%>
<br><a href="index.jsp">Back to Home</a><br><br>
<a href="viewPatients.jsp">View Patients</a>
</body></html>
```

viewPatients.jsp Code:

```
<%@ page import="java.sql.*" %>
<html><body>
<h1>Patient Records</h1>
<table border="1" cellpadding="10">
<tr>
<th>ID</th><th>Name</th><th>Age</th>
<th>Disease</th><th>Treatment Status</th><th>Actions</th>
</tr>
<%
try {
   Class.forName("com.mysql.cj.jdbc.Driver");
   Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/patientDB",
      "root",
      " PASSWORD"
   );

   Statement st = con.createStatement();
   ResultSet rs = st.executeQuery("SELECT * FROM patients");

   while(rs.next()) {
%>
<tr>
<td><%= rs.getInt("patient_id") %></td>
<td><%= rs.getString("name") %></td>
<td><%= rs.getInt("age") %></td>
<td><%= rs.getString("disease") %></td>
<td><%= rs.getString("treatment_status") %></td>
<td>
<a href="editPatient.jsp?id=<%= rs.getInt("patient_id") %>">Update</a> |
<a href="deletePatient.jsp?id=<%= rs.getInt("patient_id") %>"
onclick="return confirm('Delete this patient?')">Delete</a>
</td>
</tr>
<%
```

```
    }
    rs.close();
    st.close();
    con.close();
} catch(Exception e) {
    out.println("<tr><td colspan='6'>Error: " + e.getMessage() + "</td></tr>");
}
%>
</table>
<br><a href="index.jsp">Back to Home</a>
</body></html>
```

editPatient.jsp Code:

```
<%@ page import="java.sql.*" %>
<html><body>
<h2>Update Patient Details</h2>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/patientDB",
        "root",
        " PASSWORD"
    );

    int id = Integer.parseInt(request.getParameter("id"));
    PreparedStatement ps = con.prepareStatement(
        "SELECT * FROM patients WHERE patient_id=?"
    );
    ps.setInt(1, id);
    ResultSet rs = ps.executeQuery();

    if(rs.next()) {
%>
<form action="updatePatient.jsp" method="post">
Patient ID: <input type="number" name="patient_id"
value="<%= rs.getInt("patient_id") %>" readonly><br><br>
Name: <input type="text" name="name"
value="<%= rs.getString("name") %>" required><br><br>
Age: <input type="number" name="age"
value="<%= rs.getInt("age") %>" required><br><br>
Disease: <input type="text" name="disease"
```

```
value="<%= rs.getString("disease") %>" required><br><br>
Treatment Status:
<select name="treatment_status">
<option value="Under Treatment">Under Treatment</option>
<option value="Recovered">Recovered</option>
<option value="Discharged">Discharged</option>
</select><br><br>
<input type="submit" value="Update Patient">
</form>
<%
   }
   rs.close();
   ps.close();
   con.close();
} catch(Exception e) {
   out.println("Error: " + e.getMessage());
}
%>
</body></html>
```

updatePatient.jsp Code:

```
<%@ page import="java.sql.*" %>
<html><body>
<%
try {
   Class.forName("com.mysql.cj.jdbc.Driver");
   Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/patientDB",
      "root",
      " PASSWORD"
   );

   int id = Integer.parseInt(request.getParameter("patient_id"));

   PreparedStatement ps = con.prepareStatement(
      "UPDATE patients SET name=?, age=?, disease=?, treatment_status=?
WHERE patient_id=?"
   );

   ps.setString(1, request.getParameter("name"));
   ps.setInt(2, Integer.parseInt(request.getParameter("age")));
   ps.setString(3, request.getParameter("disease"));
```

```
    ps.setString(4, request.getParameter("treatment_status"));
    ps.setInt(5, id);

    int result = ps.executeUpdate();

    if(result > 0)
        out.println("<h2>Patient updated successfully!</h2>");
    else
        out.println("<h2>Patient not found!</h2>");

    ps.close();
    con.close();
} catch(Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
<br><a href="viewPatients.jsp">View Patients</a><br><br>
<a href="index.jsp">Back to Home</a>
</body></html>
```

deletePatient.jsp Code:

```
<%@ page import="java.sql.*" %>
<html><body>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/patientDB",
        "root",
        " PASSWORD"
    );

    int id = Integer.parseInt(request.getParameter("id"));

    PreparedStatement ps = con.prepareStatement(
        "DELETE FROM patients WHERE patient_id=?"
    );
    ps.setInt(1, id);

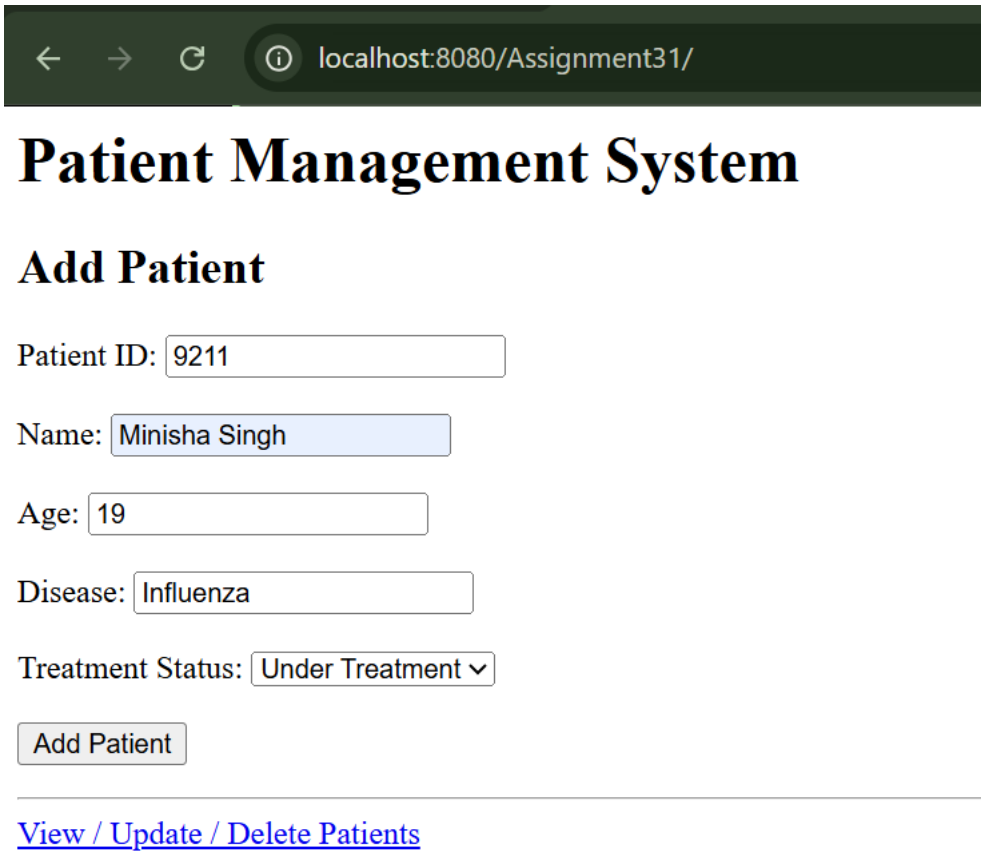
    int result = ps.executeUpdate();

    if(result > 0)
```

```
        out.println("<h2>Patient deleted successfully!</h2>");
    else
        out.println("<h2>Patient not found!</h2>");

    ps.close();
    con.close();
} catch(Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
<br><a href="viewPatients.jsp">View Patients</a><br><br>
<a href="index.jsp">Back to Home</a>
</body></html>
```

Output Screenshots:

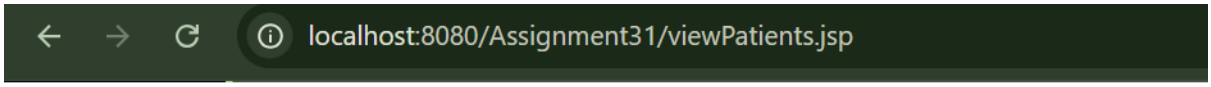




Patient added successfully!

Back to Home

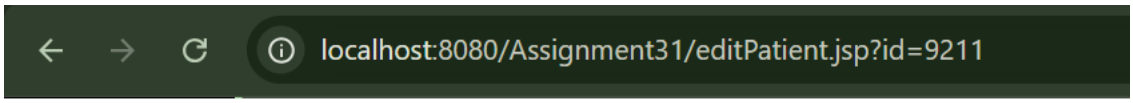
View Patients



Patient Records

ID	Name	Age	Disease	Treatment Status	Actions
9211	Minisha Singh	19	Influenza	Under Treatment	Update | Delete

Back to Home



Update Patient Details

Patient ID: 9211

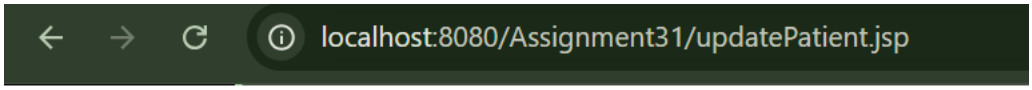
Name: Riya Singh

Age: 19

Disease: Influenza

Treatment Status: Under Treatment

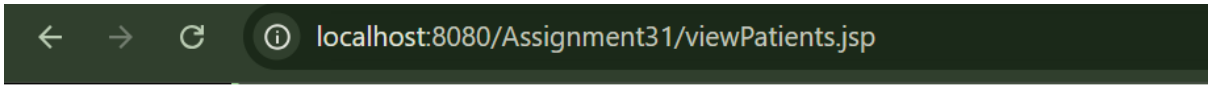
Update Patient



Patient updated successfully!

View Patients

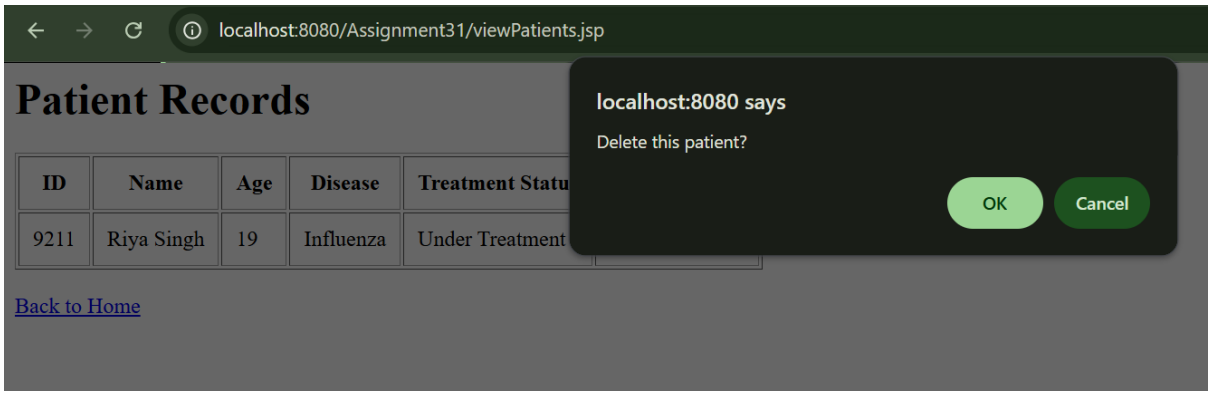
Back to Home

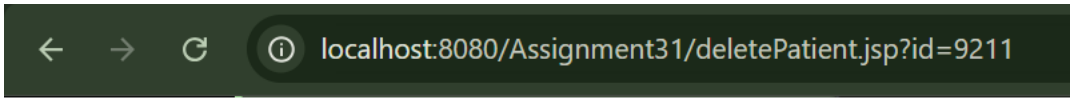


Patient Records

ID	Name	Age	Disease	Treatment Status	Actions
9211	Riya Singh	19	Influenza	Under Treatment	Update | Delete

Back to Home





Patient deleted successfully!

View Patients

Back to Home

Conclusion

Thus, the JSP-based Patient Management System was successfully implemented using CRUD operations. The application allows users to add new patient records, view existing records, update patient details, and delete records when required. MySQL was used to store and manage patient information, including patient ID, name, age, disease, and treatment status. JDBC was used to establish a connection between JSP and the database. This assignment helped in understanding database connectivity and performing CRUD operations in a web application.